

Name _____

Period _____ Date _____

Practice Test – Advanced Kinematics (Version 2)

On any problems, you must show all work, include units, round properly, and box your final answer.

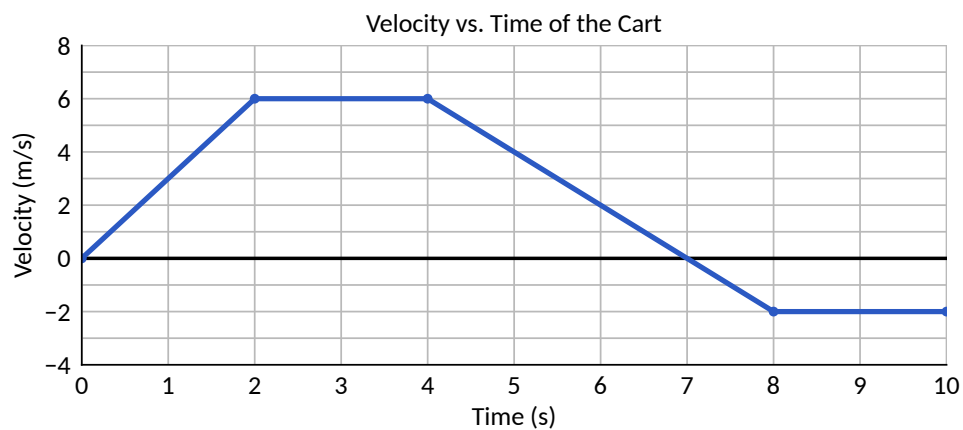
Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

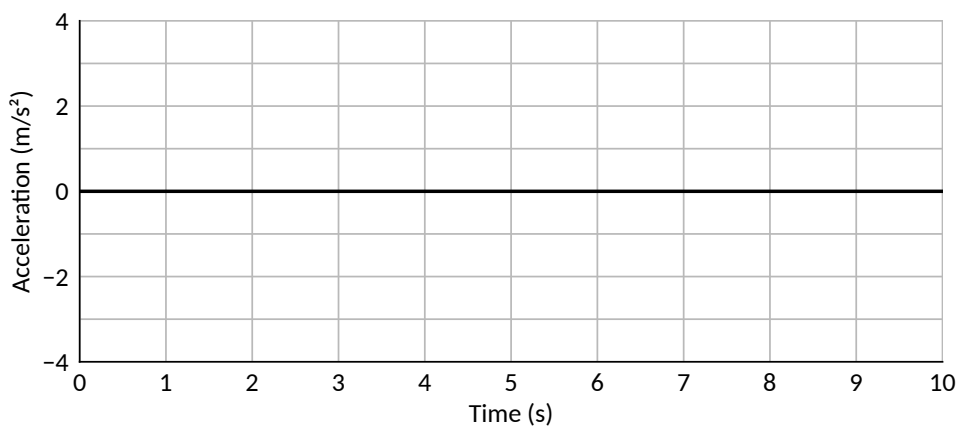
For all questions:

- Use $g = 9.80 \text{ m/s}^2$ and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) The velocity-time graph below shows the motion of a cart moving along a straight track from $t = 0$ s to $t = 10$ s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

2. (25 points) A car traveling at 27.0 m/s on a straight, level road brakes with a constant acceleration of magnitude 4.50 m/s^2 until it comes to a stop.

a. **(15 points)** How far does the car travel while braking?

b. **(5 points)** How long does it take the car to stop?

c. **(5 points)** If the car's initial speed were doubled and it braked with the same acceleration, by what factor would its braking distance change? Justify your answer using your expression from part (a).

3. (25 points) A football is kicked from ground level on a level field with an initial speed $v_0 = 22.0 \text{ m/s}$ at an angle $\theta = 40.0^\circ$ above the horizontal. It lands at the same height from which it was kicked.

a. **(15 points)** Derive an expression for the ball's range R in terms of v_0 , θ , and g . Then calculate the range.

b. **(5 points)** What is the ball's hang time?

c. **(5 points)** What is the ball's maximum height above the field?

4. (25 points) A small object is released from rest at the top of a skyscraper of height H above the ground. Air resistance is negligible. Choose upward as the positive direction.

a. **(15 points)** Derive an expression for the object's velocity just before it hits the ground, including its sign, in terms of H and g .

b. **(10 points)** Write an expression for the object's velocity $v(t)$ as a function of the time t since it was released, and state the time interval over which this expression is valid, in terms of H and g .

Answer Key – Practice Test, Version 2 – Advanced Kinematics

Fully worked. Algebra first, numbers last, units throughout. $g = 9.80 \text{ m/s}^2$; answers to three significant figures.

1. Acceleration-time graph (25 points)

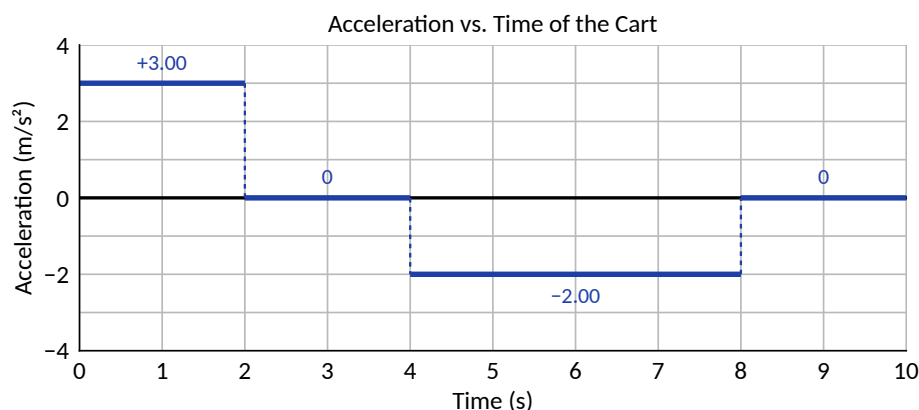
Acceleration = slope of the velocity-time graph: $a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$

$$0 \text{ s to } 2 \text{ s: } a = \frac{(6 \text{ m/s} - 0 \text{ m/s})}{(2 \text{ s} - 0 \text{ s})} = \frac{(6 \text{ m/s})}{(2 \text{ s})} = \boxed{+3.00 \text{ m/s}^2}$$

$$2 \text{ s to } 4 \text{ s: } a = \frac{(6 \text{ m/s} - 6 \text{ m/s})}{(4 \text{ s} - 2 \text{ s})} = \frac{(0 \text{ m/s})}{(2 \text{ s})} = \boxed{0 \text{ m/s}^2} \text{ (flat segment)}$$

$$4 \text{ s to } 8 \text{ s: } a = \frac{((-2 \text{ m/s}) - 6 \text{ m/s})}{(8 \text{ s} - 4 \text{ s})} = \frac{(-8 \text{ m/s})}{(4 \text{ s})} = \boxed{-2.00 \text{ m/s}^2}$$

$$8 \text{ s to } 10 \text{ s: } a = \frac{((-2 \text{ m/s}) - (-2 \text{ m/s}))}{(10 \text{ s} - 8 \text{ s})} = \frac{(0 \text{ m/s})}{(2 \text{ s})} = \boxed{0 \text{ m/s}^2} \text{ (flat segment)}$$



Each segment of constant slope on the v-t graph becomes a horizontal line on the a-t graph. The acceleration is unchanged at the instant v crosses zero.

Suggested partial credit	pts
Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each)	8
Correct slope value with sign and units for each segment (1 each)	4
Four horizontal segments at the correct accelerations (2 each)	8
Changes in acceleration occur at the correct times; no slanted lines	2
Axes labeled with quantity and units; readable scale	3
Total	25

2. Braking car (25 points)

Take the direction of motion as positive. Knowns: $v_0 = +27.0 \text{ m/s}$, $v = 0 \text{ m/s}$, $a = -4.50 \text{ m/s}^2$ (opposite the motion).

(a) Time is neither given nor asked for, so use $v^2 = v_0^2 + 2a\Delta x$.

$$0 = v_0^2 + 2a\Delta x \rightarrow \Delta x = -\frac{v_0^2}{2a}$$

$$\Delta x = -\frac{(+27.0 \text{ m/s})^2}{2(-4.50 \text{ m/s}^2)} = \frac{(729 \text{ m}^2/\text{s}^2)}{(9.00 \text{ m/s}^2)} = \boxed{81.0 \text{ m}}$$

(b) $v = v_0 + at \rightarrow 0 = v_0 + at \rightarrow t = -\frac{v_0}{a}$

$$t = -\frac{(+27.0 \text{ m/s})}{(-4.50 \text{ m/s}^2)} = \boxed{6.00 \text{ s}}$$

(c) From (a), $\Delta x = -\frac{v_0^2}{2a}$, so with the acceleration a held constant, $\Delta x \propto v_0^2$.

$$\Delta x' = -\frac{(2v_0)^2}{2a} = 4\left(-\frac{v_0^2}{2a}\right) = 4\Delta x \rightarrow \text{the braking distance is multiplied by } \boxed{4.00}$$

Check: $4(81.0 \text{ m}) = 324 \text{ m}$.

Suggested partial credit	pts
(a) Chooses $v^2 = v_0^2 + 2a\Delta x$ (time not involved)	3
(a) Correct signs: $v = 0$ and a negative (opposite the motion)	3
(a) Solves symbolically, $\Delta x = -v_0^2/(2a)$, before substituting	4
(a) Substitution with units in every parenthesis	3
(a) Final answer, 3 significant figures, units, boxed	2
(b) Correct equation solved for t , substitution with units, answer (2 + 2 + 1)	5
(c) Shows $\Delta x \propto v_0^2$ from the part (a) expression	3
(c) Factor of 4 (not 2)	2
Total	25

3. Football range (25 points)

Horizontal: $a_x = 0$, $v_{x0} = v_0 \cos \theta$. Vertical: $a_y = -g$, $v_{y0} = v_0 \sin \theta$, $\Delta y = 0$ (lands at launch height).

(a) Time of flight from the vertical: $\Delta y = v_{y0}t + \frac{1}{2}a_y t^2 \rightarrow 0 = (v_0 \sin \theta)t - \frac{1}{2}gt^2 = t(v_0 \sin \theta - \frac{1}{2}gt)$

$$t = 0 \text{ (launch)} \text{ or } t = \frac{2v_0 \sin \theta}{g}$$

$$\text{Horizontal: } R = v_{x0}t = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} = \frac{v_0^2 (2 \sin \theta \cos \theta)}{g} \rightarrow \boxed{R = \frac{v_0^2 \sin 2\theta}{g}}$$

$$R = \frac{(22.0 \text{ m/s})^2 \sin(2 \times 40.0^\circ)}{(9.80 \text{ m/s}^2)} = \frac{(484 \text{ m}^2/\text{s}^2)(\sin 80.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{48.6 \text{ m}}$$

$$\text{(b) } t = \frac{2v_0 \sin \theta}{g} = \frac{2(22.0 \text{ m/s})(\sin 40.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{2.89 \text{ s}}$$

$$\text{(c) At the top } v_y = 0: v_y^2 = v_{y0}^2 - 2g h_{\max} \rightarrow 0 = (v_0 \sin \theta)^2 - 2g h_{\max} \rightarrow h_{\max} = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$h_{\max} = \frac{(22.0 \text{ m/s})^2 (\sin 40.0^\circ)^2}{2(9.80 \text{ m/s}^2)} = \boxed{10.2 \text{ m}}$$

Check: $v_{x0} = 16.9 \text{ m/s}$ and $t = 2.89 \text{ s}$; multiplying the unrounded values gives $R = v_{x0}t = 48.6 \text{ m}$, matching part (a).

Suggested partial credit	pts
(a) Resolves v_0 into $v_0 \cos \theta$ and $v_0 \sin \theta$	2
(a) Uses $\Delta y = 0$ to get $t = 2v_0 \sin \theta / g$	4
(a) $R = (v_0 \cos \theta)t$ with the expression for t substituted (not a number)	4
(a) Simplifies to $R = v_0^2 \sin 2\theta / g$	2
(a) Numerical range with units, 3 significant figures, boxed	3
(b) $t = 2v_0 \sin \theta / g$, substitution with units, answer (2 + 2 + 1)	5
(c) Uses $v_y = 0$ at the top in $v_y^2 = v_{y0}^2 - 2g h_{\max}$	2
(c) $h_{\max} = v_0^2 \sin^2 \theta / (2g)$	1
(c) Substitution with units and answer	2
Total	25

4. Object dropped from a skyscraper (25 points), variables only

Up is positive; origin at the release point. Knowns: $v_0 = 0$, $a = -g$, displacement to the ground $\Delta y = -H$.

(a) Time is not involved, so use $v^2 = v_0^2 + 2a\Delta y$:

$$v^2 = 0^2 + 2(-g)(-H) = 2gH \rightarrow v = \pm\sqrt{2gH}$$

The object is moving downward, and down is negative, so take the negative root: $v = -\sqrt{2gH}$

Units check: $\sqrt{(m/s^2)(m)} = m/s$.

(b) $v = v_0 + at = 0 + (-g)t \rightarrow v(t) = -gt$

It is valid from release until impact. Impact time: $\Delta y = v_0 t + \frac{1}{2}at^2 \rightarrow -H = -\frac{1}{2}gt_f^2 \rightarrow t_f = \sqrt{2H/g}$

Valid interval: $0 \leq t \leq \sqrt{2H/g}$

Check: $v(t_f) = -g\sqrt{2H/g} = -\sqrt{g^2 \cdot 2H/g} = -\sqrt{2gH}$, which matches part (a).

Suggested partial credit	pts
(a) States $v_0 = 0$, $a = -g$ and $\Delta y = -H$ consistent with up positive	4
(a) Uses $v^2 = v_0^2 + 2a\Delta y$	3
(a) Algebra to $v^2 = 2gH$	3
(a) Chooses the negative root with a reason (moving down)	3
(a) Final expression $v = -\sqrt{2gH}$, boxed	2
(b) $v(t) = -gt$ from $v = v_0 + at$	4
(b) Impact time $t = \sqrt{2H/g}$ found from $-H = -\frac{1}{2}gt^2$	3
(b) Interval $0 \leq t \leq \sqrt{2H/g}$ stated	2
(b) Consistency with part (a) (or other check)	1
Total	25